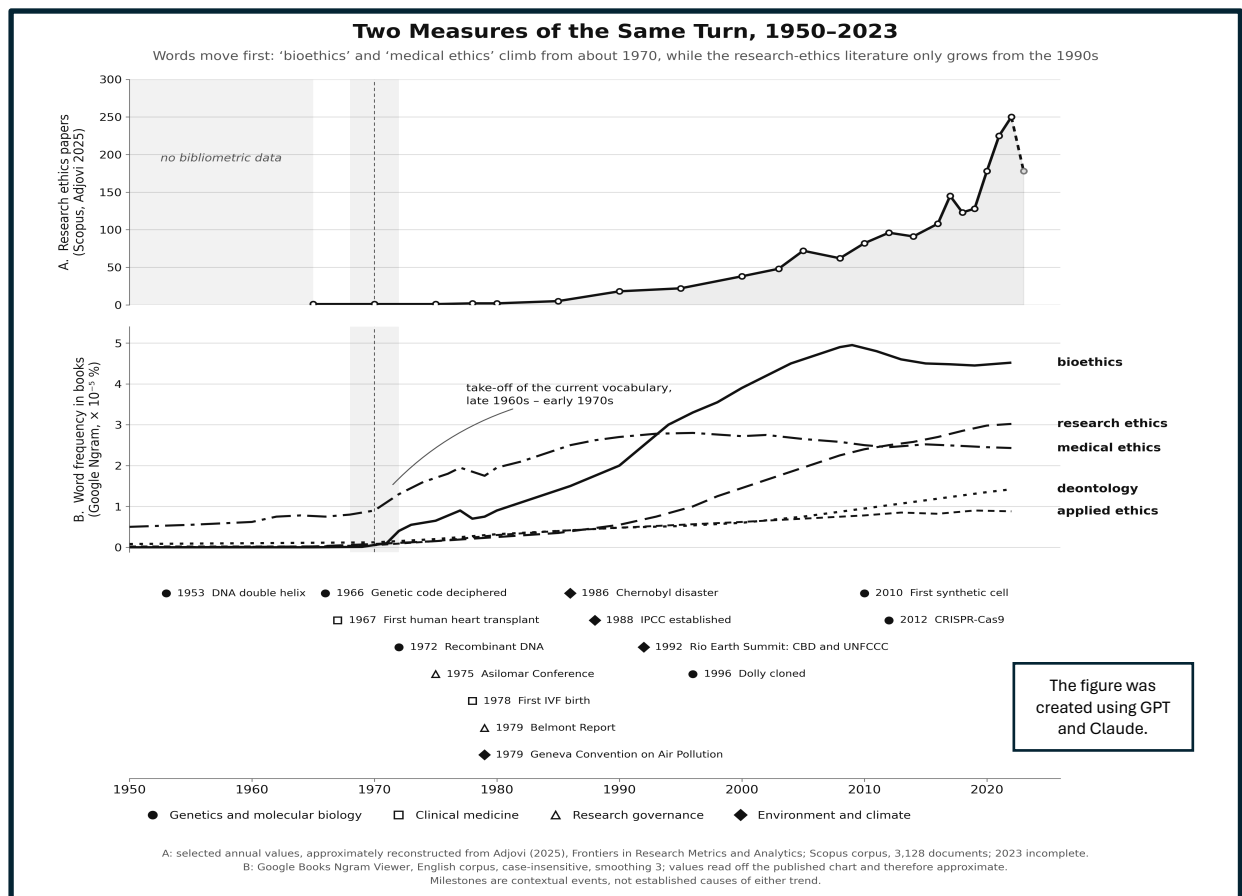


Putting Morality to the Test: Ethical Principles and Their Consequences



The figure sets two independent traces of the same historical episode side by side. *Panel A* counts research-ethics papers indexed in Scopus, reconstructed from Adjovi's 2025 bibliometric study of 3,128 documents; *panel B* tracks how often five words appear in the English-language books digitized by Google, expressed as a share of all words. The first measures a research field's institutional output, the second the diffusion of a vocabulary - and the two do not move together. Words move first. The frequencies of *medical ethics* and, more sharply, *bioethics* begin to climb around 1970, the latter from near zero to roughly a fivefold share by the late 2000s; *research ethics*, *deontology* and *applied ethics* follow at greater distance. The bibliometric curve, by contrast, stays essentially flat until the early 1990s and only turns steep after 2000, with its sharpest rise in the last decade. Between the moment a language becomes available and the moment a literature organizes itself around it, the figure suggests a lag of roughly two decades. This is what one would expect if the vocabulary were the condition rather than the product of the field: authors need a name for what they are doing before journals, committees and funding lines can be built around it. Consider that the two panels are not commensurable. Moreover, word frequency in books measures uptake by writers and publishers, not by clinicians, regulators or patients: it tells us when a language became sayable, not when it began to do work.

The dated markers along the axis are contextual, not causal. They are placed to let the reader hold two chronologies together - the discovery of the double helix, the first heart transplant, recombinant DNA, Asilomar, Chernobyl, IVF, Rio Earth Summit, Dolly, etc. - and to ask whether the take-off of the current vocabulary in the late 1960s and early 1970s coincides with the arrival of scientific knowledge, technical capacities and environmental problems that made older moral categories unusable. Nothing in the figure demonstrates that it does.

When science and ethics are discussed, four assumptions are usually taken for granted:

- **Science and technology are devoid of any intrinsic ethics** and, to recognize the limits of their choices and actions, must turn to ideas developed within philosophical, humanistic, and/or religious traditions. In other words, freedom and rights, the good, human dignity, justice, and moral conscience are not constitutive elements of science and scientific practice but belong exclusively to the humanistic dimensions of ethics. This assumption also underlies the idea that recourse to ethics—particularly in the education and training of researchers—serves to cleanse science of its problematic features and to orient it toward the good.
- **Scientific and empirical knowledge is entirely irrelevant to ethical argumentation**, and it is considered fallacious to appeal to scientific facts to derive or construct a value.
- **From the first two assumptions follows the idea that, by looking within oneself and embracing the “right” ethics, one can understand what it is good or right to decide or do.** This understanding is assumed to be independent of any objective verification of the facts. The good and the right are, in this view, already within us.
- **Science and ethics are often treated as two epistemologically homogeneous worlds that guide the production of arguments.** Neither “science” nor “ethics” exists as an abstract, unified entity. What exist are individuals who conduct empirical research or examine choices from an ethical point of view. These individuals adopt specific methodologies and organize themselves within academic institutions or public and private research centers. Such institutions interact with one another and with other social and political institutions, through processes of discussion and negotiation that ultimately influence political and regulatory decision-making.

These are some of the misconceptions we see at work when scientists say that “we need an ethics to guide our choices in research and innovation,” or when scholars in the humanities argue that “science has no ethics of its own, does not explain everything, and does not fit within a shared cultural framework.” It is true that “science does not think,” as a philosopher with a highly problematic political biography once put it; but neither ethics nor philosophy thinks. Only individuals think.

Scientists and humanists nevertheless tend to agree that ethics, law, and politics are necessary because “not everything that can be done ought to be done,” and because the “values widely shared in society” need to be negotiated and translated into limits on science and innovation. There is nothing wrong with maintaining that, in liberal democracies, law and politics constitute, at least in principle, the ultimate arena in which the demands and implications of research and innovation are governed. However, without controlled data and empirically corroborated explanations, law, politics, and ethics themselves risk producing decisions that are largely arbitrary or ideological.

The lecture will begin with four cases that are particularly representative of contemporary debates and controversies concerning science and ethics, together with the questions they raise about the principles and values that should guide decision-making in such contexts. More specifically, the cases concern AI-driven synthetic biology; scientific misconduct and the deterioration of scientific publishing; the consequences of the growing role of *offshore clinical research* in drug development; and, finally, the way in which the psychometrics of risk enters political decision-making concerning the dangerousness of chemical products.

The next step will be to state—without attempting a full demonstration, for obvious reasons of time—my position on ethics, its nature, its relationship with the human moral sense, and the role played by science in the evolution of the value framework of Western ethics. Ethics does not exist in an abstract, uniform world independent of historical and cultural contexts. Rather, it emerges within cultures from social emotions, which are subsequently processed through cognitive mechanisms that, under natural conditions, are inevitably subject to biases. Quite apart from the fact that this cultural and cognitive dimension deserves to be much more widely understood, it has an important implication: ethical reasoning can easily be influenced by non-rational factors, including intuitions, emotional salience, stereotypes, and mental representations that may be irrelevant to the actual structure of the problem being considered.

Science, understood as an epistemologically structured practice of knowledge production, emerged in its modern form in the seventeenth century. Since then, researchers have continuously had to negotiate the conditions, limits, and implications of scientific inquiry with existing systems of values, political and religious institutions, and economic and productive systems. The lecture will therefore proceed by examining the broad historical development of these processes of negotiation. The present relationship between ethics and science should not be regarded as a stable or definitive condition. The forms we observe today, as well as those toward which contemporary debate appears to be moving, need to be understood from a historical perspective. Without this historical depth, we risk overlooking the constitutive features and recurring patterns of the phenomena we are trying to understand. The conditions under which research is conducted, the questions addressed to science, social expectations, ethical and social criteria of evaluation, and the role of institutions have all progressively changed.

Historical and cultural knowledge does not allow us to determine automatically what ought to be done, nor does it enable us to predict precisely what will happen. Understanding the historical depth of a phenomenon does not generate normative prescriptions. It can, however, perform an essential function: it can make us more cautious, more aware of the complexity of the problems we face, and less inclined to trust those who offer rigid, universal, and ready-made solutions. Four centuries of interaction among science, ethics, institutions, and society have produced the contemporary landscape, which can be represented through the following framework:

Snapshot of a Debate

The ethics of science is conceived as having the function of setting limits on research - that is, prohibitions: post-Enlightenment ethics places the emphasis on the research freedom to test solutions while respecting persons and their living contexts. Ethics should safeguard the production of reliable knowledge, the minimization of harm, and openness to criticism.

THREE LEVELS OF RESPONSIBILITY, HABITUALLY CONFLATED

The researcher: honesty, method, declared conflicts of interest

Institutions: review, incentives, correction

Applications: the province of politics

WHEN RESEARCH STOPS

Epistemic / material limits

Cognitive capacities, instruments, available theory, funding, biological timescales, reproducibility, the complexity of systems.
The most frequent. Not ethical in kind.

Non-negotiable ethical limits

Autonomy and informed consent. The prohibition against causing suffering. The prohibition of coercive experimentation.
Non-negotiable.

Negotiable moral limits

Embryos, gene editing, animal experimentation, artificial intelligence.
They shift with social values.

Natural limits

Biological immortality, zero risk, perfect health, the total eradication of infections.
Impossibilities, not prohibitions

The freedom of research is not indiscriminate: selection operates through methodological quality, plausibility, reproducibility, and opportunity cost — not through the moralization of its object.

THREE RECURRING DISTORTIONS OF THE DEBATE

Sacralization — Embryo, genome, nature, biodiversity, species — treated as natural moral categories rather than as constructs with a cultural history.

Asymmetric precaution — The risks of acting are weighed, never those of failing to act. The principle makes sense only if applied to both sides.

The demand for certainty — Results are provisional and probabilistic. Honesty consists in communicating uncertainty, not in suppressing it.

The contemporary paradox: science is asked to solve enormous problems and at the same time distrusted; it is regulated ever more tightly, held to a standard of absolute certainty, and its results are moralized.

Scientific advances and technological innovations are continuously reshaping the context, giving rise to new ethical challenges and requiring the development of appropriate approaches to address them.